

Workshop 3 - 02bash

Parte 1. Primer servicio Nginx

```
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ touch docker-compose.yml

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ touch docker-compose.yml

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ ls
docker-compose.yml

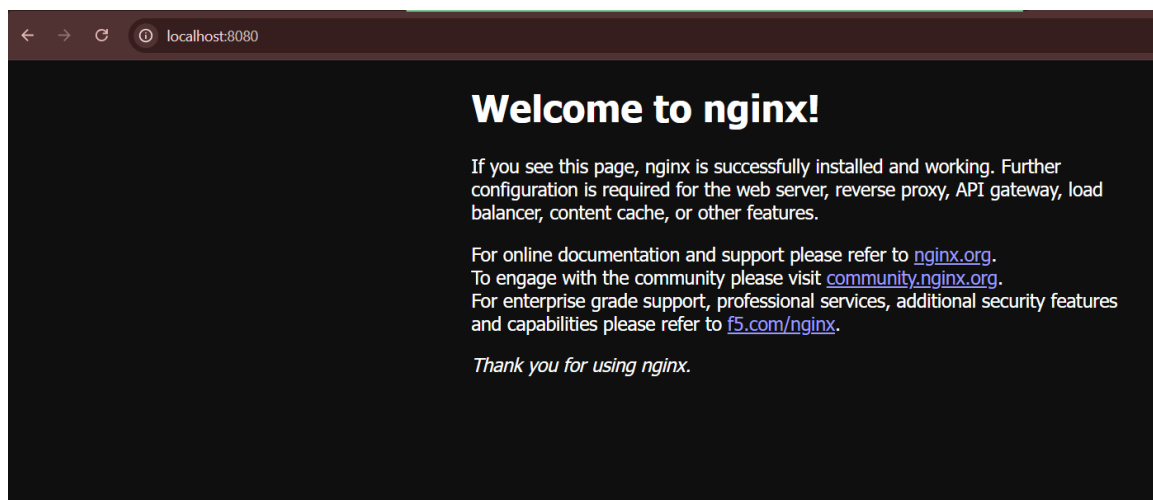
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ nano docker-compose.yml

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ cat docker-compose.yml
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"  GNU nano 8.2
```

```
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ docker compose up -d
Image nginx:alpine Pulling
fb597529c916 Pulling fs layer 0B
1ed8e39a7434 Pulling fs layer 0B
0d62d88506ba Pulling fs layer 0B
da4dea1b00af Pulling fs layer 0B
55afa1ecc21d Pulling fs layer 0B
d94291c26261 Pulling fs layer 0B
0ea935727878 Pulling fs layer 0B
8cdfef4f23778 Pulling fs layer 0B
0d62d88506ba Downloading 404B
da4dea1b00af Download complete 0B
76cd5dde9375 Download complete 0B
8cdfef4f23778 Download complete 0B
0d62d88506ba Download complete 0B
0ea935727878 Download complete 0B
43aeb74b2469 Download complete 0B
1ed8e39a7434 Download complete 0B
fb597529c916 Downloading 1.049MB
fb597529c916 Downloading 1.049MB
d94291c26261 Downloading 1.902MB
fb597529c916 Downloading 2.097MB
fb597529c916 Downloading 2.097MB
d94291c26261 Downloading 1.902MB
fb597529c916 Downloading 3.146MB
```

```
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ docker compose ps
NAME          IMAGE          COMMAND                  SERVICE    CREATED      STATUS      PORTS
ws3-web-1    nginx:alpine   "/docker-entrypoint..." web         31 seconds ago Up 29 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
```



```

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ docker compose logs web
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/08/30 19:30:21 [notice] 1#1: using the "epoll" event method
web-1 | 2026/08/30 19:30:21 [notice] 1#1: nginx/1.31.4
web-1 | 2026/08/30 19:30:21 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
web-1 | 2026/08/30 19:30:21 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
web-1 | 2026/08/30 19:30:21 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker processes
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 31
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 32
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 33
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 34
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 35
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 36
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 37
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 38
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 39
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 40
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 41
web-1 | 2026/08/30 19:30:21 [notice] 1#1: start worker process 42
web-1 | 172.18.0.1 - - [30/Aug/2026:19:31:06 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT
ebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
web-1 | 2026/08/30 19:31:06 [error] 31#31: *2 open() "/usr/share/nginx/html/favicon.ico" failed (2: No suc
ient: 172.18.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer
/web-1 | 172.18.0.1 - - [30/Aug/2026:19:31:06 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:
dows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"

```

```

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ docker compose down
Container ws3-web-1 Stopping
Container ws3-web-1 Stopped
Container ws3-web-1 Removing
Container ws3-web-1 Removed
Network ws3_default Removing
Network ws3_default Removed

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ docker compose ps
NAME                IMAGE                                COMMAND                  SERVICE   CREATED      STATUS      PORTS

```

Parte 2. Nginx con contenido propio

```
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ mkdir html

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ ls
docker-compose.yml  html/

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ touch html/index.html

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ ls html
index.html
```

```
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ cat > html/index.html << 'EOF'
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <title>Mi primer servicio Docker</title>
</head>
<body>
  <h1>Hola desde Nginx y Docker Compose</h1>
  <p>Este contenido está siendo servido desde un contenedor.</p>
</body>
</html>
EOF

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ cat > docker-compose.yml << 'EOF'
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html:ro
EOF

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$
```



Hola desde Nginx y Docker Compose

Este contenido está siendo servido desde un contenedor.

Preguntas

1. ¿Por qué el cambio aparece sin reconstruir la imagen?

Porque la carpeta `html` del host está montada directamente dentro del contenedor mediante un volumen. Nginx lee directamente los archivos de esa carpeta, por lo que los cambios realizados en el host se reflejan sin reconstruir la imagen.

2. ¿Qué función cumple `:ro` ?

`ro` significa **read-only**, es decir, solo lectura.

El contenedor puede leer los archivos de `html`, pero no debería modificarlos.

3. ¿Cuál es la diferencia entre el puerto `8080` y el `80` ?

`8080` es el puerto del **host**, es decir, de nuestro computador.

`80` es el puerto dentro del **contenedor**, donde Nginx está escuchando.

4. ¿Qué parte representa el host y qué parte representa el contenedor?

En:

```
"8080:80"
```

tenemos:

```
8080 = host  
80   = contenedor
```

Parte 3. Compose utilizando un Dockerfile

```
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)  
$ touch Dockerfile  
  
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)  
$ ls  
Dockerfile  docker-compose.yml  html/
```

```
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)  
$ nano Dockerfile  
  
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)  
$ cat Dockerfile  
FROM nginx:alpine  
COPY html/ /usr/share/nginx/html/  
  
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)  
$
```

```
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)  
$ nano docker-compose.yml  
  
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)  
$ cat docker-compose.yml  
services:  
  web:  
    build: .  
    ports:  
      - "8080:80"
```

```

- "8080:80"

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ docker compose up --build -d
Image ws3-web Building
#1 [internal] load local bake definitions
#1 reading from stdin 493B 0.0s done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 89B 0.0s done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/nginx:alpine
#3 DONE 0.3s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 330B done
#5 DONE 0.1s

#6 [1/2] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120
288c127ffaeeb4a6d4b95a26fead017d5913
#6 resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288
c127ffaeeb4a6d4b95a26fead017d5913 0.0s done
#6 DONE 0.2s

#7 [2/2] COPY html/ /usr/share/nginx/html/
#7 DONE 0.0s

#8 exporting to image
#8 exporting layers 0.1s done
#8 exporting manifest sha256:9d1de4fe5e35b37dc61aaf42a3d10a53208f293ee1c982eaad7
f37cc81eec595 0.0s done
#8 exporting config sha256:6ae25d36e5b8c4ade47ad43a58a91305e258e17c3b56c2617be94
b49616d0cc3 0.0s done
#8 exporting attestation manifest sha256:c0e5cad09e7fe14b49e6ae75ae1b0c69850e7b6
e0cccacc90199e0b88124e44 0.0s done
#8 exporting manifest list sha256:33dbf2d93e9bc2c3e5efeb152e9722a127f7057208e273
e4b36d32b6ea17e8ff 0.0s done
#8 naming to docker.io/library/ws3-web:latest
#8 exporting sha256:9d1de4fe5e35b37dc61aaf42a3d10a53208f293ee1c982eaad7f37cc81eec595
0.0s done
#8 DONE 0.1s

```

```

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ docker compose ps
NAME          IMAGE          COMMAND                  SERVICE    CREATED      STATUS      PORTS
ws3-web-1    ws3-web        "/docker-entrypoint..." web        27 seconds ago Up 26 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)

```

Parte 4. Reto integrador

```

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ mkdir -p ubuntu scripts logs

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ ls
Dockerfile  docker-compose.yml  html/  logs/  scripts/  ubuntu/

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ touch scripts/monitor.sh

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ nano scripts/monitor.sh

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ cat scripts/monitor.sh
#!/bin/bash

echo "Monitor iniciado"

while true
do
    echo "$(date) - Monitor ejecutándose" >> /logs/monitor.log
    sleep 10
done

```

```

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ touch scripts/analiza.sh

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ nano scripts/analiza.sh

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ cat scripts/analiza.sh
#!/bin/bash

echo "Analizando logs..."

if [ -f /logs/monitor.log ]
then
    echo "Número de líneas:"
    wc -l /logs/monitor.log

    echo "Últimas líneas:"
    tail /logs/monitor.log
else
    echo "No existe el archivo monitor.log"
fi

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ |

```

```

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ chmod +x scripts/monitor.sh

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ chmod +x scripts/analiza.sh

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ ls -l scripts
total 2
-rwxr-xr-x 1 feijo 197609 241 Aug 30 15:16 analiza.sh*
-rwxr-xr-x 1 feijo 197609 134 Aug 30 15:16 monitor.sh*

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$

```

```

#19 [ubuntu-service 6/6] RUN chmod +x monitor.sh analiza.sh
#19 CACHED

#20 [ubuntu-service] exporting to image
#20 exporting layers done
#20 exporting manifest sha256:255f580dcf27985177d8885eca16fca201438992d46d60e74e9eb9c88c7eca2a done
#20 exporting config sha256:c3dcb95ae3811390057ba1617c4e65d63635823cff74d8252fd87df295eb1d1d done
#20 exporting attestation manifest sha256:7afa1c9cd8df1c1271fffc77406b935621bf1c2cdf13806fce9c56f3eeca6fa8 0.0s done
#20 exporting manifest list sha256:038fcaa40d749380a2a9b7e64de016cffd57117de47dff9397d1abd186b6d80b
#20 exporting manifest list sha256:038fcaa40d749380a2a9b7e64de016cffd57117de47dff9397d1abd186b6d80b 0.0s done
#20 naming to docker.io/library/ws3-ubuntu-service:latest done
#20 unpacking to docker.io/library/ws3-ubuntu-service:latest done
#20 DONE 0.1s

#21 [ubuntu-service] resolving provenance for metadata file
#21 DONE 0.0s
Image ws3-web Built
Image ws3-ubuntu-service Built
Network ws3_default Creating
Network ws3_default Created
Container ws3-web-1 Creating
Container ws3-ubuntu-service-1 Creating
Container ws3-ubuntu-service-1 Created
Container ws3-web-1 Created
Container ws3-ubuntu-service-1 Starting
Container ws3-web-1 Starting
Container ws3-web-1 Started
Container ws3-ubuntu-service-1 Started

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$ docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
ws3-ubuntu-service-1	ws3-ubuntu-service	"tail -f /dev/null"	ubuntu-service	8 seconds ago	Up 7 seconds	
ws3-web-1	ws3-web	"/docker-entrypoint..."	web	8 seconds ago	Up 7 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp

```

feijo@Valenfei MINGW64 ~/02-bash/WS3 (master)
$

```



Hola desde Nginx y Docker Compose

Este contenido está siendo servido desde un contenedor.

Pregunta

Docker Compose es más conveniente porque permite definir todos los servicios, puertos, volúmenes, imágenes, redes y configuraciones de una aplicación en un único archivo `docker-compose.yml`. De esta manera, varios contenedores pueden iniciarse, detenerse y administrarse conjuntamente mediante comandos como `docker compose up` y `docker compose down`, en lugar de ejecutar y configurar manualmente múltiples comandos `docker run`.